Aeolus Renewables

Drivetrain condition assessment using 10-minute SCADA measurements

131,760
sensor records

15
turbines

3.00%
recorded faults

Predictive Modeling with Machine Learning and Neural Networks | Low-Code Path

Submission date: October 11, 2026

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Executive Summary

VALIDATION RECALL	TEST RECALL	TEST PRECISION	TEST F1
92.49%	**86.40%**	**49.48%**	**62.93%**
Tuned ANN selected	1,245 / 1,441 faults	False-alert tradeoff	Balance of P and R

The operational decision

Use the ANN as analyst decision support, not an automatic turbine shutdown trigger.

1,271 false-positive records require remote triage; 196 fault records were missed.

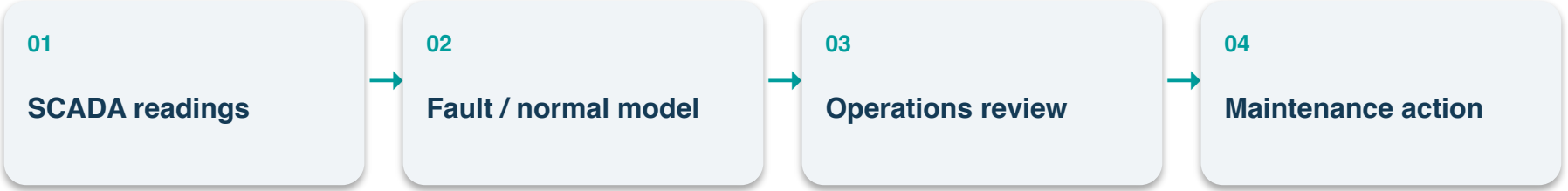
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Business Problem Overview and Solution Approach

WHY IT MATTERS

Missed drivetrain faults can lead to extended downtime, emergency repairs and lost energy production.

CONDITION-BASED WORKFLOW

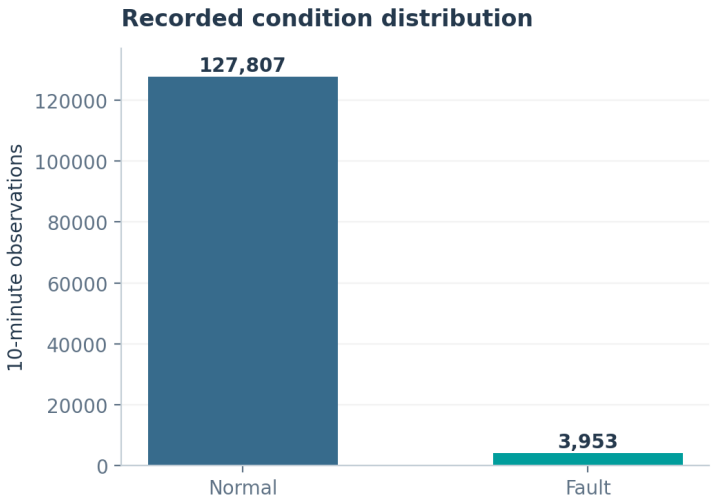


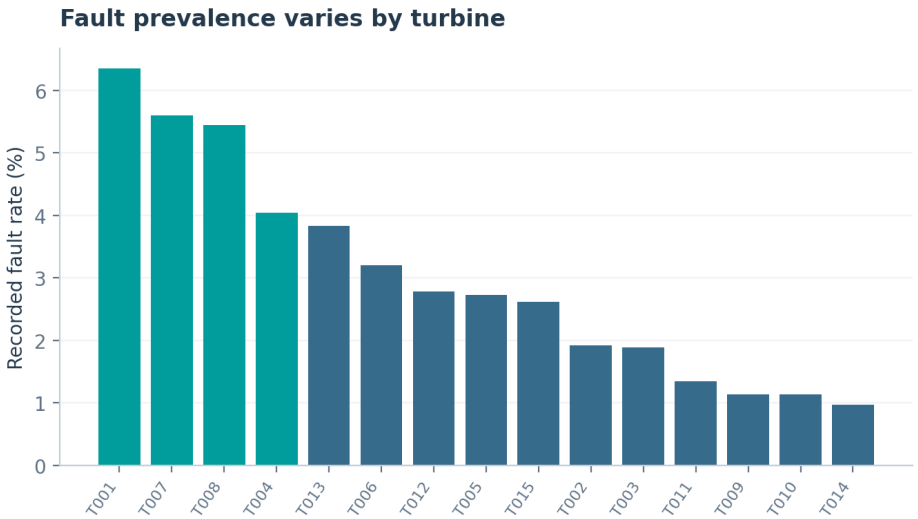
Method: timestamp-grouped chronological split, training-only preprocessing, five baseline models, two tuned candidates, validation-driven selection.

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EDA Results — Class Balance and Asset Patterns

131,760 readings • 15 turbines • 3.00% recorded faults

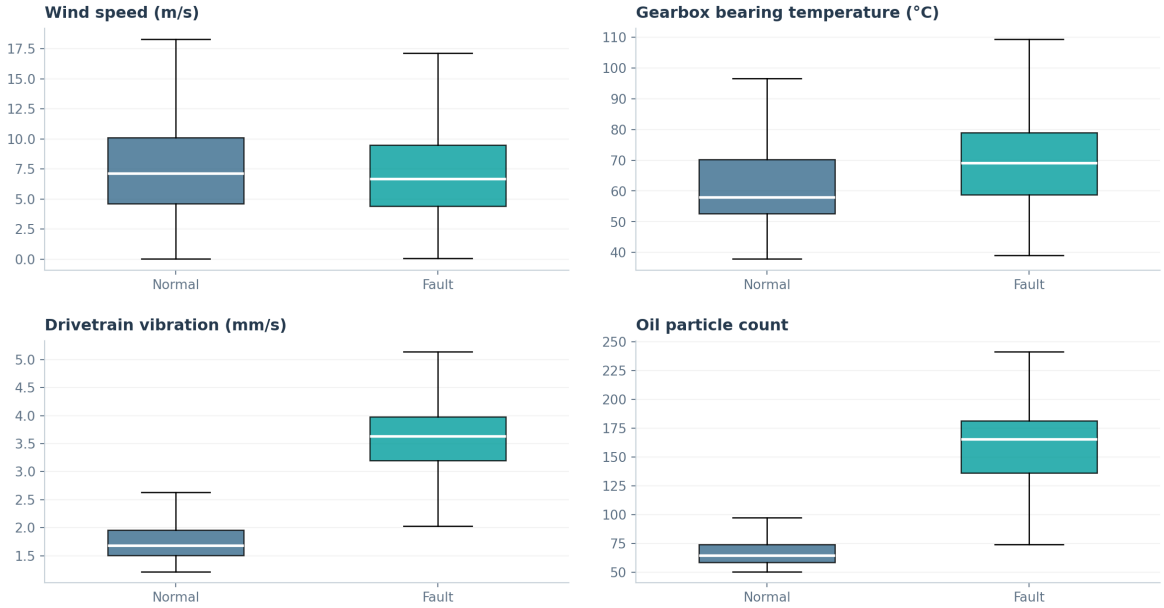




The minority fault class makes accuracy alone misleading; turbine-level rates support prioritised investigation.

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EDA Results — Sensor Evidence



KEY OBSERVATIONS

Vibration, gearbox temperature and oil particles are visibly elevated in fault-state records.

Wind speed alone provides weaker separation, supporting multi-signal evaluation.

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Data Preprocessing and Leakage Prevention

DATA QUALITY

0 duplicate rows; missing gearbox-oil (645), generator-bearing (690), oil-pressure (636).

TREATMENT

Training-median imputation; ANN StandardScaler fit using training rows only.

FEATURE ENGINEERING

Time cycles, power utilisation and temperature-rise variables.

EVALUATION DESIGN

Timestamp-grouped chronological 70% / 15% / 15%; test reserved until final evaluation.

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Model Performance Summary and Final Selection

Tuned model	Validation recall	F1	Precision
ANN (selected)	92.49%	53.54%	37.67%
XGBoost	69.10%	69.34%	69.57%

Selection rule

Maximise validation recall; F1 and precision are secondary tie-breakers.

ANN catches more recorded fault states. XGBoost offers better precision and F1, so the false-alert tradeoff must be managed.

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Actionable Business Recommendations

01 Triage

Review persistent model alerts and corroborating sensor signals remotely.

02 Dispatch

Prioritise technician inspections when risk persists across successive intervals.

03 Learn

Link alerts to work orders; track false alarms, missed faults, and operational workload.

04 Validate

Run prospective field validation before promising lead time or cost savings.

Evidence base: 196 missed SCADA fault observations support recall-first selection. The 1,271 false-positive SCADA observations are not 1,271 technician visits or incidents; remote triage and alert persistence should determine dispatch. Pilot KPIs: missed observations, alert workload, work-order confirmation, downtime and drift.

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SCADA panel

8,784 timestamp groups, recorded at 10-minute intervals across 15 turbines.

Environment

Wind, ambient conditions

Operation

Power and rotational speeds

Condition

Bearing heat, vibration, oil particles

Asset history

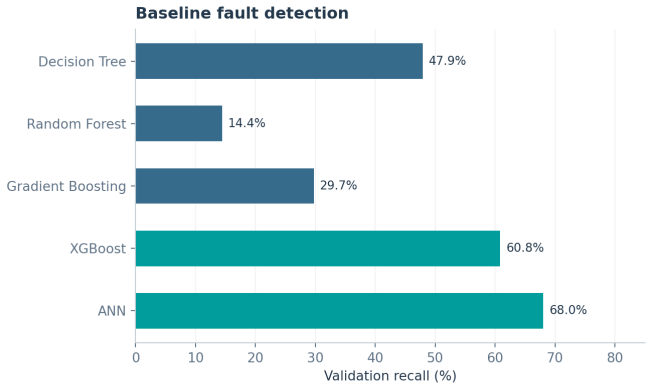
Hours, load cycles and maintenance

Target: failure = 1 denotes a recorded fault condition, not a future failure horizon.

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Model Building — Five Baseline Classifiers

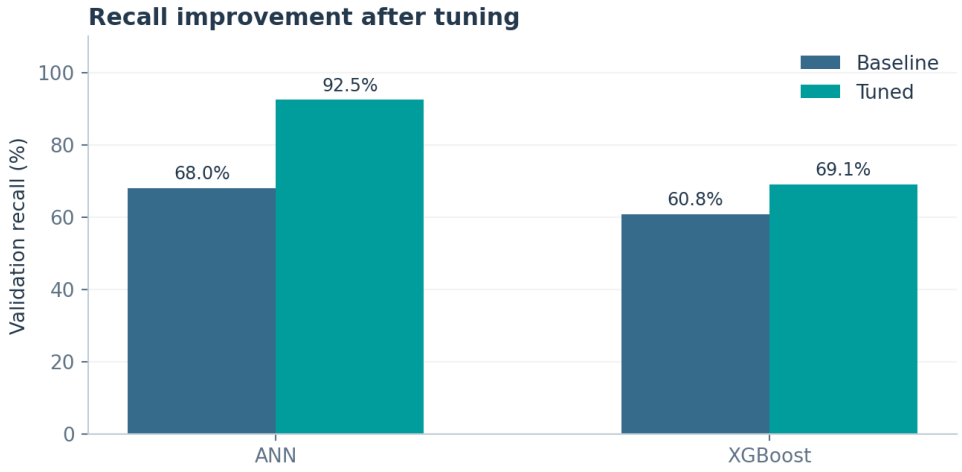
Model	Accuracy	Precision	Recall	F1
Decision Tree	90.70%	27.74%	47.93%	35.14%
Random Forest	95.35%	82.87%	14.44%	24.59%
Gradient Boosting	94.54%	46.96%	29.74%	36.42%
XGBoost	96.64%	71.01%	60.83%	65.53%
ANN	94.79%	50.28%	68.05%	57.83%



ANN and XGBoost have the highest baseline recall and were selected for tuning.

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Model Improvement — Hyperparameter Tuning



XGBOOST

Six randomized XGBoost configurations; two timestamp-grouped expanding-window folds; recall objective.

Selected XGBoost: 160 trees, depth 3, learning rate 0.07, subsample 0.85, colsample 1.00, min-child weight 3. Best CV recall 48.45%; holdout validation recall 69.10%.

ANN

ANN: 10 seed configurations were evaluated on the fixed timestamp-grouped chronological validation partition: baseline reproduction, prior 64/24 regularised ANN, wider L2, higher-dropout L2, and compact regularised. Median imputation and scaling were fit on training data only; training-label class weights and validation-loss early stopping were used. The recorded prior 64/24 configuration retained the highest validation recall (92.49%); best new run was higher-dropout L2 at 92.40%.

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Final Model Performance — Untouched Test

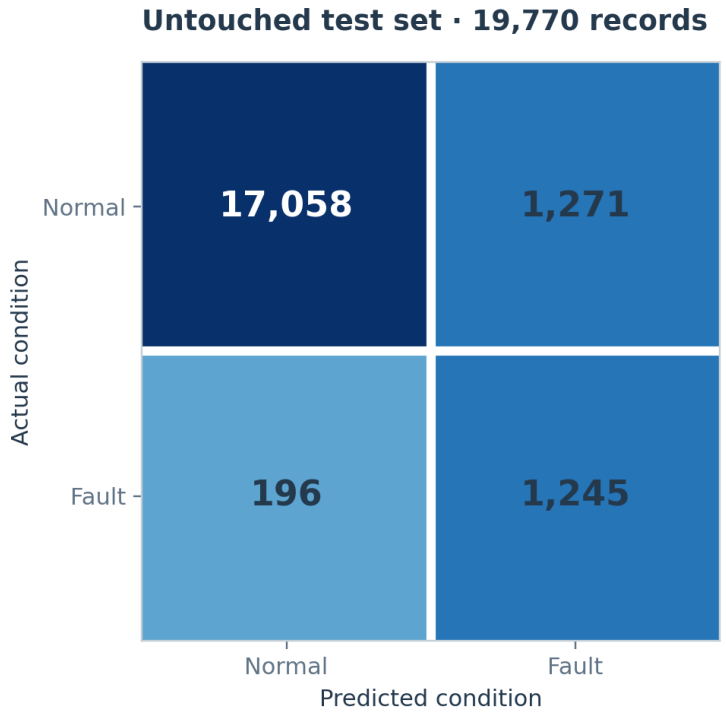
TEST RECALL
86.40%

TEST PRECISION
49.48%

TEST ACCURACY
92.58%

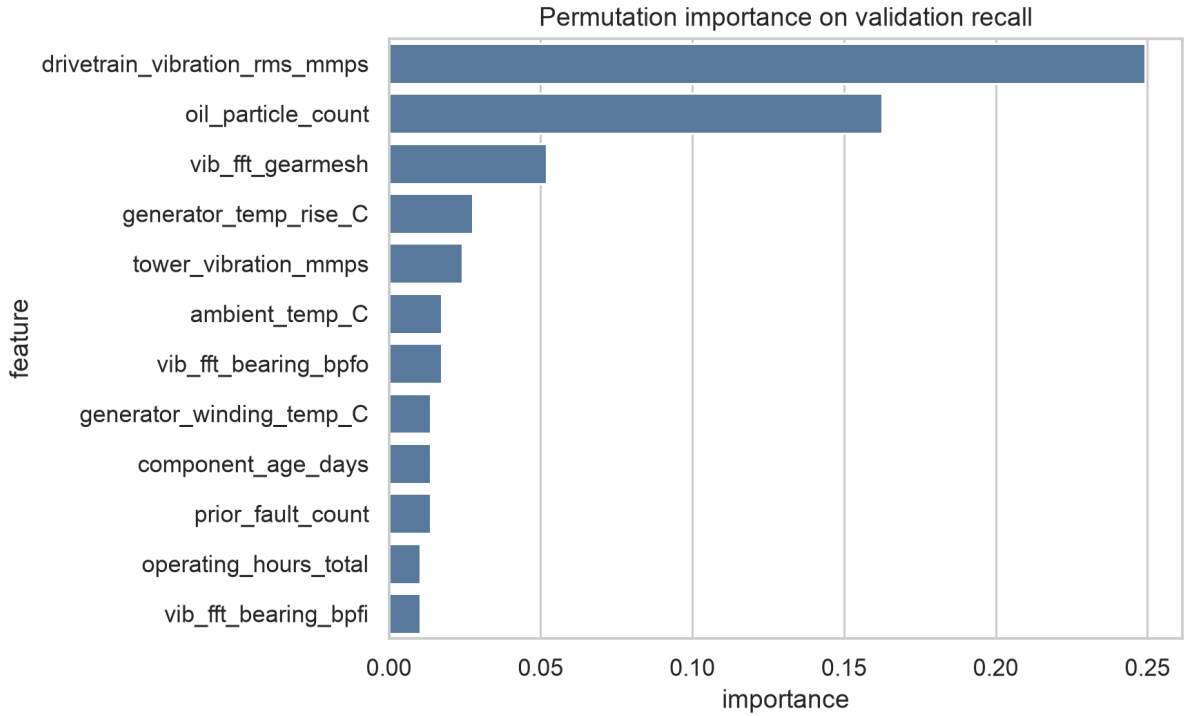
TEST F1
62.93%

1,245 detected • 196 missed
1,271 false alerts • 17,058 true normals



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Feature Interpretation and Limitations



KEY DRIVERS

Drivetrain vibration and oil particle count show the largest validation-recall impacts.

Permutation importance measures association, not causation or a future-failure lead time.

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Timestamp-Grouped Split Verification

TRAIN

92,220

6,148 timestamps

VALIDATION

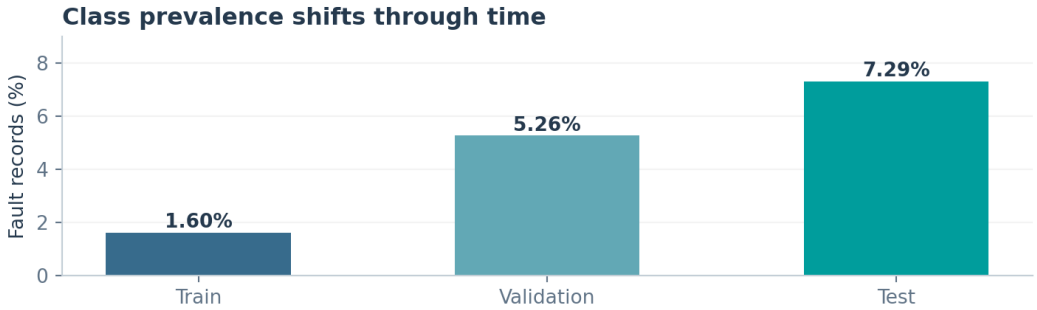
19,770

1,318 timestamps

TEST

19,770

1,318 timestamps



LEAKAGE CHECK

All 15 turbine records at a timestamp remain in the same partition. Failure prevalence changes over time.

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Operational Use Case and Next Steps

STAGE 1

Model flags current abnormal condition.

STAGE 2

Analyst checks persistence and supporting sensors.

STAGE 3

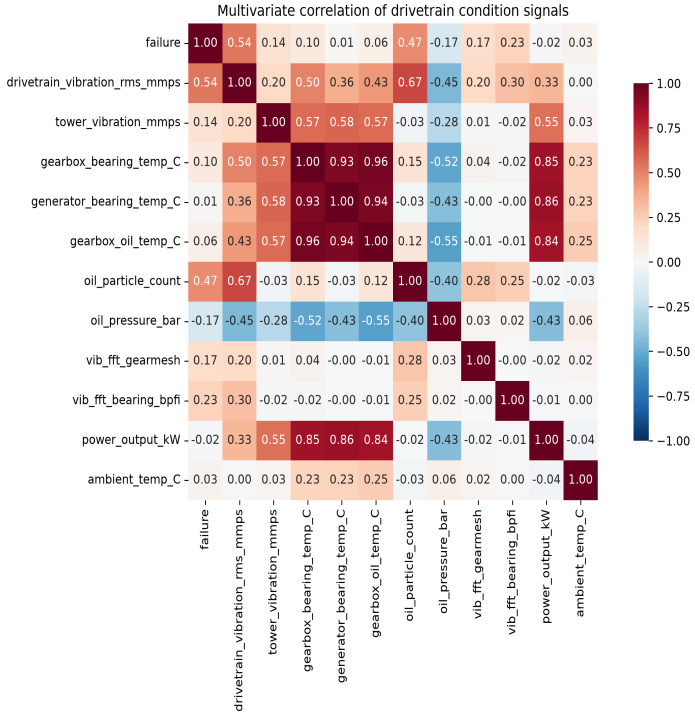
Maintenance team prioritises planned inspection.

DEPLOYMENT GUARDRAILS

Retrospective proof of concept: prospectively validate alert value, technician workload, model drift and downtime outcomes before automated action or savings claims.

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EDA Results — Multivariate Sensor Relationships



What the heatmap adds

- Drivetrain vibration has the strongest correlation with recorded fault state (0.54).
- Oil-particle count is also material (0.47); bearing-frequency vibration is 0.23.
- Oil pressure is negative (−0.17); gearbox-bearing temperature is weaker (0.10).

Implication: Fault-state evidence is distributed across signals. A multi-feature model is more appropriate than a single sensor threshold.

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Tuning Comparison — Recall First, Precision Trade-off

Candidates were selected from baseline validation recall. Test data did not influence tuning, selection or threshold choice.

Model	Validation P / R / F1	Status	Configuration / evidence
ANN	37.67 / 92.49 / 53.54	Selected	Recorded 64/24; BatchNorm; dropout 0.30/0.15; Adam; early stopping
ANN	45.13 / 92.40 / 60.64	Best new run	Higher-dropout L2; 5 actual fixed-seed ANN configurations evaluated
XGBoost	69.57 / 69.10 / 69.34	Tuned comparator	6 randomized configurations; 2 timestamp-grouped expanding folds

Selection rationale
Validation recall is primary. The historical 64/24 ANN remains selected at 92.49%, narrowly above the best fresh configuration (92.40%). Its lower precision/F1 reflects the alert trade-off. XGBoost has higher precision and F1 but lower recall (69.10%). Test data did not choose configurations or thresholds.

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